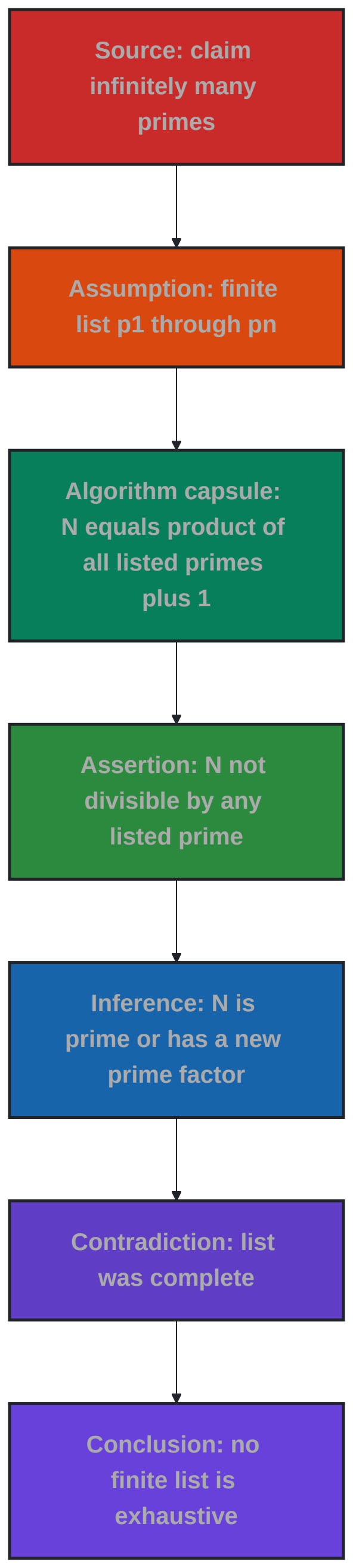


**Source: claim
infinitely many
primes**



```
graph TD; A[Source: claim infinitely many primes] --> B[Assumption: finite list p1 through pn]; B --> C[Algorithm capsule: N equals product of all listed primes plus 1]; C --> D[Assertion: N not divisible by any listed prime]; D --> E[Inference: N is prime or has a new prime factor]; E --> F[Contradiction: list was complete]; F --> G[Conclusion: no finite list is exhaustive];
```

**Assumption: finite
list p_1 through p_n**

**Algorithm capsule:
 N equals product of
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**Assertion: N not
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**Conclusion: no
finite list is
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